

Answer Key & Teacher Notes

Session 1 – Equations and Solutions • Session 2 – Writing Equations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Equations and Solutions

TALK PROMPTS

- Why can an equation be false for one value of the variable and true for another?
- Two partners tested different values and got different results. How can both checks still be correct work?

PARTNER TASK

Form A, problem 1a. The equation is $s - 15 = 25$. One partner tests $s = 40$; the other tests $s = 35$. Each shows the substitution, then together they decide which value is the solution and why.

WATCH FOR

- Reporting the value of each side (42) as the solution instead of the value of the variable (12).
- Reading $4p$ as $4 + p$ instead of $4 \times p$.
- Accepting a value that makes the two sides “close” but not equal.

SESSION 2 • TEACHER NOTES

Writing Equations

TALK PROMPTS

- Why is a good estimate a better starting point than a random number?
- Why does your first estimate not have to be right for this method to work?

PARTNER TASK

Coat Drive. A drive collected c coats on Friday and 37 more on Saturday, 82 in all. One partner writes the equation; the other estimates and tests one value. Switch roles and revise until both sides match.

WATCH FOR

- Writing the parts in the wrong order for a take-away story ($72 + m = 25$ or $18 - b = 60$).
- Writing $9 - n$ for “a number decreased by 9.”
- Using $+$ for an equal-groups story instead of $\times$.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.B.5	1	2	2	1	1	2	2	3	3	1	1
S1 • Form B	6.EE.B.5	1	2	2	1	1	2	2	3	3	1	1
S2 • Form A	6.EE.B.6	1	2	2	2	2	2	2	3	3	1	1
S2 • Form B	6.EE.B.6	1	2	2	2	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

equation: A math sentence that says two amounts are **equal**.

variable: A letter that stands for an **unknown** number.

solution: A value that makes the equation **true**.

substitute: **Replace** the variable with a number.

- To test a value, **substitute** it for the variable and do the math.
- If both sides are **equal**, the value is a solution. If not, it is not a solution.

Example: $k = 7$: $7 - 4 = 3$. $3 \neq 2$, so the equation is **false**.
 $k = 6$: $6 - 4 = 2$. $2 = 2$, so the equation is true. Answer: $k = 6$ is the solution.

YOU TRY

- a. T ($20 = 20$) b. F ($2 \neq 6$)
- $m = 8$, because $7 \times 8 = 56$

EXIT TICKET

- a. T ($19 = 19$) b. F ($18 \neq 21$)
- $h = 8$, because $8 - 5 = 3$
- A** (value)

PRACTICE FORM A

- a. T ($25 = 25$) b. F ($35 \neq 42$) c. T ($47 = 47$) d. F ($128 \neq 36$) e. T ($3 = 3$) f. T ($41 = 41$)
- a. $21 + 12 = 33$, true b. $19 + 12 = 31 \neq 33$, false c. $j = 21$
- $p = 12$, because $6 \times 12 = 72$
- $60 - 24 = 36$, so yes, $c = 60$ is a solution.
- $4 \times 7 = 28$, so yes, Ray is correct.
- $f - 2 = 11$; $f = 13$, because $13 - 2 = 11$
- $5 \times 8 = 40 \neq 45$, so $g = 8$ is not a solution. $g = 9$, because $5 \times 9 = 45$.
- a. $n + 17 = 45$ b. $28 + 17 = 45$, true, so $n = 28$ is the solution c. Testing checks one given value; solving finds the value that makes the equation true.
- Her check is right, but the solution is the value of the variable that makes the equation true. The solution is $k = 12$; 42 is the value of each side.
- A** (yes)
- B** (no)

Self-check word **EQUAL**

PRACTICE FORM B

- a. T ($30 = 30$) b. F ($54 \neq 48$) c. T ($51 = 51$) d. F ($72 \neq 27$) e. T ($4 = 4$) f. T ($44 = 44$)
- a. $27 + 14 = 41$, true b. $25 + 14 = 39 \neq 41$, false c. $j = 27$
- $p = 13$, because $8 \times 13 = 104$
- $65 - 18 = 47$, so yes, $c = 65$ is a solution.
- $5 \times 6 = 30 \neq 35$, so no, Ray is not correct ($t = 7$).
- $f - 3 = 14$; $f = 17$, because $17 - 3 = 14$
- $7 \times 9 = 63 \neq 56$, so $g = 9$ is not a solution. $g = 8$, because $7 \times 8 = 56$.
- a. $n + 19 = 52$ b. $33 + 19 = 52$, true, so $n = 33$ is the solution c. Testing checks one given value; solving finds the value that makes the equation true.
- $40 \neq 45$, so $x = 15$ is not a solution. Both sides must be exactly equal. The solution is $x = 20$, because $20 + 25 = 45$.
- A** (yes)
- B** (no)

Self-check word **VALUE**

GUIDED NOTES

variable: A letter that stands for the **unknown** number.

equation: A math sentence with an **equal** sign.

increased by: Means **add**.

decreased by: Means **subtract**.

- Name the unknown with a **letter** and say what it means.
- Write the story as a math sentence, like start – lost = left, or groups $\times$ size = **total**.

Example: Let m = the marbles that **rolled away**. Start – lost = left: $30 - m = 11$. Test $m = 19$: $30 - 19 = 11$. True. Answer: **19** marbles rolled away.

YOU TRY

- $n + 8 = 15$ b. $3n = 21$
- $46 - c = 13$; $c = 33$ cartons ($46 - 33 = 13$)

EXIT TICKET

- $n + 14 = 31$ b. $4n = 36$
- $50 - b = 32$; $b = 18$ books
- B** ($8w = 56$)

PRACTICE FORM A

- $n + 17 = 40$ b. $6n = 54$ c. $n - 8 = 19$ d. $n \div 3 = 7$
- $s + 28 = 61$ b. $s = 33$, because $33 + 28 = 61$
- $m - 15 = 39$ b. $50 - 15 = 35 \neq 39$, too small; $m = 54$, because $54 - 15 = 39$
- $44 - g = 26$; $g = 18$ students
- $58 + w = 90$; $w = 32$ liters
- $c - 75 = 22$; $c = 97$ cans, because $97 - 75 = 22$
- $45b = 180$ b. $45 \times 4 = 180$, true, so $b = 4$ buses
- a. Let s = the number of students; $120 + 8s = 360$ b. $120 + 8 \times 30 = 120 + 240 = 360$, so yes c. $8s = 360$ ($s = 45$)
- Removing means subtract, so the equation should be $72 - m = 25$. His answer $m = 47$ is right: $72 - 47 = 25$. ($72 + 47$ is not 25.)
- B** ($n + 12 = 30$)
- D** ($n - 9 = 14$)

Self-check word TOTAL

PRACTICE FORM B

- $n + 23 = 50$ b. $8n = 72$ c. $n - 11 = 25$ d. $n \div 6 = 5$
- $c + 37 = 82$ b. $c = 45$, because $45 + 37 = 82$
- $m - 12 = 44$ b. $50 - 12 = 38 \neq 44$, too small; $m = 56$, because $56 - 12 = 44$
- $52 - g = 35$; $g = 17$ students
- $47 + w = 80$; $w = 33$ liters
- $c - 64 = 29$; $c = 93$ cans, because $93 - 64 = 29$
- $40b = 160$ b. $40 \times 4 = 160$, true, so $b = 4$ buses
- a. Let s = the number of students; $90 + 6s = 270$ b. $90 + 6 \times 30 = 90 + 180 = 270$, so yes c. $6s = 270$ ($s = 45$)
- The starting amount comes first: $60 - b = 18$. Her answer $b = 42$ is right: $60 - 42 = 18$. ($18 - 42$ is not 60.)
- C** ($7n = 63$)
- D** ($n - 6 = 21$)

Self-check word SOLVE

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • EQUATIONS AND SOLUTIONS

1. a. T ($20 = 20$) b. F ($30 \neq 35$) c. T ($16 = 16$) d. F ($8 \neq 7$)
e. T ($30 = 30$) f. T ($54 = 54$) g. T ($4 = 4$) h. F ($72 \neq 63$)
2. $x = 25$. Checks: $15 + 16 = 31$ (no); $25 + 16 = 41$ (yes);
 $35 + 16 = 51$ (no).
3. $b = 11$: $11 - 7 = 4$, and $4 \neq 18$, so no. $b = 25$: $25 - 7 = 18$, so yes. The solution is $b = 25$.
4. $4p$ means $4 \times p$, not $4 + p$. Substituting gives $4 \times 8 = 32$, and $32 \neq 12$, so $p = 8$ is not a solution. The solution is $p = 3$, because $4 \times 3 = 12$.
5. $t = 36$: $6 \times 36 = 216$, and $216 \neq 42$, so no. $t = 7$: $6 \times 7 = 42$, so yes. The solution is $t = 7$ tickets.

SHEET 2 • WRITING EQUATIONS

1. a. $n + 9 = 31$ b. $7n = 63$ c. $n - 12 = 20$ d. $n \div 5 = 8$
2. $s - 14 = 29$. Test: $43 - 14 = 29$, so $s = 43$ is the solution.
3. $4v = 56$. Test 12: $4 \times 12 = 48$ (no). Test 14: $4 \times 14 = 56$ (yes). $v = 14$ students.
4. $5p = 85$. Test: $5 \times 17 = 85$, so $p = 17$ pages a day is the solution.
5. Omar's equation gives $b = 17$, which would make the brother older than Tia. The brother's age plus 4 is Tia's age: $b + 4 = 13$, so $b = 9$.
6. **B** ($9w = 72$)
7. **C** ($n = 42$)